

Decent Coaching Classes

X Semi

Mind Your Math - VII

Marks: 30

Time: 1 Hr

Instructions: All questions are compulsory (internal choice as indicated). Figures to the right indicate full marks.

Q.1 (A) Choose the correct alternative.

[4]

- (1) For the equations $2x + y = 5$ and $x - y = 1$, the value of D is _____.
(A) -3 (B) 3 (C) -1 (D) 1
- (2) The roots of the equation $x^2 - 5x + 6 = 0$ are _____.
(A) 2 and 3 (B) -2 and -3 (C) 1 and 6 (D) -1 and -6
- (3) The common difference of the A.P. 10, 7, 4, 1, ... is _____.
(A) 3 (B) -3 (C) 7 (D) -7
- (4) The 10th term of the A.P. 1, 3, 5, 7, ... is _____.
(A) 17 (B) 19 (C) 21 (D) 20

Q.1 (B) Solve the following.

[4]

- (1) Write the equation $3x^2 = 5x - 2$ in the standard form $ax^2 + bx + c = 0$.
- (2) Find the value of the discriminant ($b^2 - 4ac$) of $x^2 - 3x + 2 = 0$.
- (3) Is the sequence 1, 4, 9, 16, ... an A.P.? Give a reason.
- (4) Write the first three terms of the A.P. whose first term is 7 and common difference is 3.

Q.2 (A) Complete any two of the following activities.

[4]

- (1) Solve by Cramer's rule: $x + y = 4$; $2x - y = 2$
 $D = 1 \times (-1) - 1 \times 2 = \underline{\hspace{2cm}}$, $D_x = 4 \times (-1) - 1 \times 2 = \underline{\hspace{2cm}}$, $D_y = 1 \times 2 - 4 \times 2 = \underline{\hspace{2cm}}$
 $x = D_x/D = \underline{\hspace{2cm}}$, $y = D_y/D = \underline{\hspace{2cm}}$
- (2) Solve by factorisation: $x^2 - 7x + 12 = 0$
 $x^2 - \underline{\hspace{2cm}} - 4x + 12 = 0$
 $x(x - \underline{\hspace{2cm}}) - 4(x - \underline{\hspace{2cm}}) = 0$
 $(x - 3)(\underline{\hspace{2cm}}) = 0$, $\therefore x = \underline{\hspace{2cm}}$ or $x = \underline{\hspace{2cm}}$
- (3) Find the 10th term of the A.P. 12, 16, 20, 24, ...
Here $a = \underline{\hspace{2cm}}$, $d = t_2 - t_1 = 16 - 12 = \underline{\hspace{2cm}}$, $n = 10$. $t_n = a + (n - 1)d$
 $\therefore t_{10} = \underline{\hspace{2cm}} + (10 - 1) \times \underline{\hspace{2cm}} = \underline{\hspace{2cm}} + \underline{\hspace{2cm}} = \underline{\hspace{2cm}}$

Q.2 (B) Solve any two of the following.

[4]

- (1) Solve by elimination method: $x + y = 7$; $2x - y = 5$
- (2) Find the discriminant of $x^2 - 6x + 9 = 0$ and hence write the nature of its roots.
- (3) Find the 12th term of the A.P. 3, 8, 13, 18, ...

Q.3 (A) Complete any one of the following activities.

[3]

- (1) Solve using the formula: $x^2 + 10x + 9 = 0$
Here $a = 1$, $b = \underline{\hspace{2cm}}$, $c = \underline{\hspace{2cm}}$. $b^2 - 4ac = \underline{\hspace{2cm}} - 36 = \underline{\hspace{2cm}}$
 $x = (-b \pm \sqrt{b^2 - 4ac}) / 2a = (-10 \pm \underline{\hspace{2cm}}) / 2$

$$\therefore x = \underline{\hspace{2cm}} \text{ or } x = \underline{\hspace{2cm}}$$

- (2) Which term of the A.P. 4, 9, 14, ... is 79?

Here $a = 4$, $d = \underline{\hspace{2cm}}$, $t_n = 79$. $t_n = a + (n - 1)d$

$$79 = 4 + (n - 1) \times \underline{\hspace{2cm}}, \therefore 75 = 5(n - 1), \therefore n - 1 = \underline{\hspace{2cm}}$$

$$\therefore n = \underline{\hspace{2cm}}. 79 \text{ is the } \underline{\hspace{2cm}} \text{ term.}$$

Q.3 (B) Solve any two of the following.

[6]

- (1) Solve by Cramer's rule: $3x + y = 7$; $x + 2y = 4$
- (2) Solve by factorisation: $2x^2 + 5x - 3 = 0$
- (3) Find the 20th term of the A.P. 7, 11, 15, 19, ...

Q.4 Solve any one of the following.

[5]

- (1) The cost of 2 books and 3 pens is ₹90, and the cost of 3 books and 2 pens is ₹110. Find the cost of one book and of one pen.
- (2) The product of two consecutive positive integers is 132. Find the integers.
- (3) A girl saves ₹100 in the first month, ₹150 in the second month, ₹200 in the third month and so on. Find (i) her saving in the 10th month (ii) in which month she will save ₹800.